```
clc; clear; close all;
% Tristin Webb
% Final Project
% Date: 05/5/2026

fprintf('********************************\n')
fprintf('*          Final Project       *\n')
fprintf('********************************\n')

********************************
*          Final Project       *
********************************
```

Voice-Code Breaker

```
================================================

fprintf('\n=========================================')
fprintf("\nI) Voice-Code Breaker:\n");
fprintf('=========================================\n')


%  1) Load corrupted voice data file
corrupted_file = 'Tristin_Webb.wav';
[voiceData, fs] = audioread(corrupted_file);

% Display the info for corrupted file
fprintf("\nCorrupted Audio File Information:\n")
fprintf('================================\n')
info = audioinfo(corrupted_file);
disp(info);

% Find number of digits in corrupted file
num_of_digits = info.TotalSamples / 16000;
fprintf("Number of digits in corrupted file:");
fprintf('\n================================\n');
fprintf("%i\n", num_of_digits);

% Load template files
[template_0, fs] = audioread("Zero.wav");
[template_1, fs] = audioread("One.wav");
[template_2, fs] = audioread("Two.wav");
[template_3, fs] = audioread("Three.wav");
[template_4, fs] = audioread("Four.wav");
[template_5, fs] = audioread("Five.wav");
[template_6, fs] = audioread("Six.wav");
[template_7, fs] = audioread("Seven.wav");
[template_8, fs] = audioread("Eight.wav");
[template_9, fs] = audioread("Nine.wav");

% Store these templates into an array/list
templates = [template_0, template_1, template_2, template_3, template_4,
```

```
template_5, template_6, template_7, template_8, template_9];

% First digit of corrupt file
first_corrupt_digit = voiceData(1:16000, 1);

% Second digit of corrupt file
second_corrupt_digit = voiceData(16001:32000, 1);

% Third digit of corrupt file
third_corrupt_digit = voiceData(32001:48000, 1);

% Fourth digit of corrupt file
fourth_corrupt_digit = voiceData(48001:64000, 1);


% Find the magnitude of the inner product of each digit and the templates
% Using the template and the corrupted file
% % Note: Choose the inner product with the greatest magnitude




% Find magnitude of inner product (first digit)
% ======================================================================
InnerProduct_first_digit = [1:10];
for i = 1:10
    InnerProduct_first_digit(i) = abs(dot(first_corrupt_digit,
templates(:,i)));
end

% Find index where the max inner product occurs (first digit)
maxMagnitudeFirstDigit = max(InnerProduct_first_digit);
index_first_digit = find(InnerProduct_first_digit == maxMagnitudeFirstDigit);
first_identified_digit = templates(:, index_first_digit);
% sound(first_identified_digit, fs);
% ======================================================================




% Find magnitude ofinner product (second digit)
% ======================================================================
InnerProduct_second_digit = [1:10];
for i = 1:10
    InnerProduct_second_digit(i) = abs(dot(second_corrupt_digit,
templates(:,i)));
end

% Find index where the max inner product occurs (second digit)
maxIndexSecondDigit = max(InnerProduct_second_digit);
index_second_digit = find(InnerProduct_second_digit == maxIndexSecondDigit);
```

```
second_identified_digit = templates(:, index_second_digit);
% sound(second_identified_digit, fs);
% =========================================================================




% Find magnitude of inner product (third digit)
% =========================================================================
InnerProduct_third_digit = [1:10];
for i = 1:10
    InnerProduct_third_digit(i) = abs(dot(third_corrupt_digit,
templates(:,i)));
end

% Find index where the max inner product occurs (third digit)
maxIndexThirdDigit = max(InnerProduct_third_digit);
index_third_digit = find(InnerProduct_third_digit == maxIndexThirdDigit);
third_identified_digit = templates(:, index_third_digit);
% sound(third_identified_digit, fs);
% =========================================================================




% Find magnitude of inner product (fourth digit)
% =========================================================================
InnerProduct_fourth_digit = [1:10];
for i = 1:10
    InnerProduct_fourth_digit(i) = abs(dot(fourth_corrupt_digit,
templates(:,i)));
end

% Find index where the max inner product occurs (fourth digit)
maxIndexFourthDigit = max(InnerProduct_fourth_digit);
index_fourth_digit = find(InnerProduct_fourth_digit == maxIndexFourthDigit);
fourth_identified_digit = templates(:, index_fourth_digit);
% sound(fourth_identified_digit, fs);
% =========================================================================



% Display decoded voice-code numbers and the magnitude of inner-products
fprintf("\nDecoded Voice-Code Numbers:\n");
fprintf("==================================\n");
fprintf("First Number: %d (Magnitude: %.2f)\n", index_first_digit-1,
maxMagnitudeFirstDigit);
fprintf("Second Number: %d (Magnitude: %.2f)\n", index_second_digit-1,
maxIndexSecondDigit);
fprintf("Third Number: %d (Magnitude: %.2f)\n", index_third_digit-1,
```

```
maxIndexThirdDigit);
fprintf("Fourth Number: %d (Magnitude: %.2f)\n", index_fourth_digit-1,
maxIndexFourthDigit);


% Combine the identified digits into a single audio signal
recovered_audio = [first_identified_digit; second_identified_digit;
third_identified_digit; fourth_identified_digit];

% Write the combined audio signal to a file
audiowrite('recovered_audio.wav', recovered_audio, fs);

% Audio player object to listen to recovered audio
player = audioplayer(recovered_audio, fs);
player.play();



% Plot the plot for first 1000 samples of first digit and another plot for
% first 1000 samples of the corresponding digit that was identified

% Plot the first 1000 samples of the first digit and the corresponding
identified digit
figure;
subplot(2, 1, 1);
plot(first_corrupt_digit(1:1000));
title('First Corrupt Digit');
xlabel('Sample');
ylabel('Amplitude');

% Plot the first 1000 samples of the identified digit
subplot(2, 1, 2);
plot(first_identified_digit(1:1000));
title('First Identified Digit');
xlabel('Sample');
ylabel('Amplitude');


% Extra Credit:
% Write code so that it could decode a number of any length
% ========================================================================

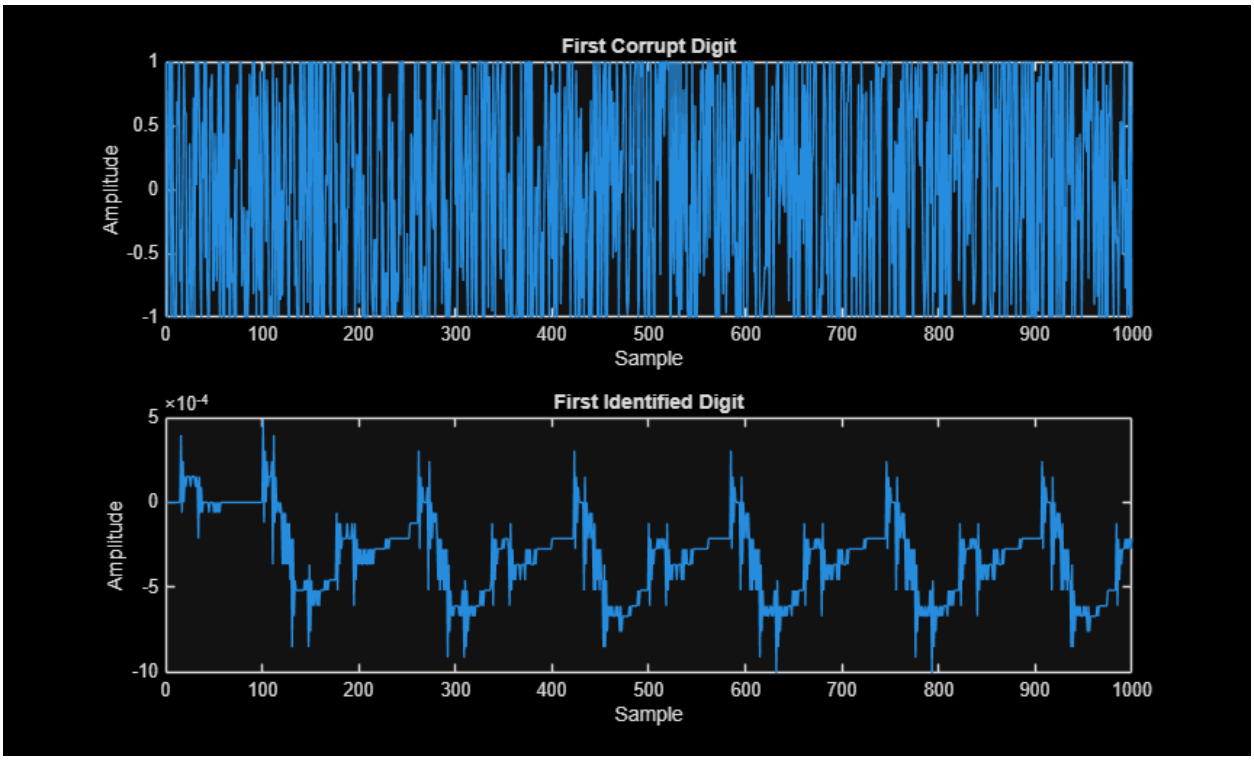

========================================
I) Voice-Code Breaker:
========================================

Corrupted Audio File Information:
================================
             Filename:
'C:\Users\lunix\OneDrive\Desktop\EE200_Final_Proj\Tristin_Webb.wav'
    CompressionMethod: 'Uncompressed'
          NumChannels: 1
           SampleRate: 16000
```

```
          TotalSamples: 64000
              Duration: 4
                 Title: []
               Comment: []
                Artist: []
          BitsPerSample: 16

Number of digits in corrupted file:
================================
4

Decoded Voice-Code Numbers:
================================
First Number: 9 (Magnitude: 45.06)
Second Number: 8 (Magnitude: 27.26)
Third Number: 5 (Magnitude: 33.92)
Fourth Number: 0 (Magnitude: 59.74)
```



Design of an experiment

```
================================================

fprintf('\n=========================================')
fprintf("\nII) Design of an experiment:\n");
fprintf('=========================================\n')

% Random values modeled as Gaussian
std_dev = 10; % ohms
```

```
% Find the mean value of the resistors produced by taking some samples
% Use those samples to estimate the mean value



% a) 95% confident that the true mean is present within (+/-) 4 ohms of
%    your estimated mean.... How will you design this experiment?
%
% How many sample resistors n need to be measured to estimate the mean
% value.
% ========================================================================

confidence_level = 0.95; % c
critical_value = 1.960;  % z*
margin_of_error = 4; % m

% Margin of error formula: (z*)*(std_dev/sqrt(n)) = m
% z* = critical value
% n = # of samples
% m = margin of error

% Solve for n

n = ceil(((std_dev * critical_value)/margin_of_error)^2);


fprintf("\nSamples needed to estimate the mean value:\n");
fprintf('================================\n');
fprintf("%i\n", n);




% b) Later the manager tells you the true mean value is 100 ohm. Generate n
%    random resistor values to simulate the above scenario and estimate the
%    mean resistor value.
%
% What is the confidence interval? Does your confidence interval contain
% the true mean?
% ========================================================================



% Generate random resistor values centered around the true mean
true_mean = 100; % ohms
sample_resistors = true_mean + std_dev * randn(n, 1);

% Find the sample mean
sample_mean = mean(sample_resistors);

% Determine the confidence interval
```

```
margin_of_error = critical_value * (std_dev/sqrt(n));
lower = sample_mean - margin_of_error;
upper = sample_mean + margin_of_error;

fprintf("\nConfidence Interval:\n");
fprintf('================================\n');
fprintf("[%.2f, %.2f]\n", lower, upper);

if (true_mean >= lower && true_mean <= upper)
    fprintf("Confidence Interval contains true mean\n");
else
    fprintf("Confidence Interval does NOT contain true mean\n");
end




% c) Repeat part b 1000 times (do not print out all the resistor values)
%
% Note: Do not report each mean estimate nor the confidence intervals. Only
% need to report how many times, out of 1000 repetitions, is the true mean
% contained within the confidence interval?
% ========================================================================

num_trials = 1000;
count_within_interval = 0;

lower_limits = zeros(num_trials,1);
upper_limits = zeros(num_trials,1);

for i = 1:num_trials
    sample = true_mean + std_dev * randn(n,1);
    sample_mean = mean(sample);

    lower = sample_mean - margin_of_error;
    upper = sample_mean + margin_of_error;

    lower_limits(i) = lower;
    upper_limits(i) = upper;

    if (true_mean >= lower && true_mean <= upper)
        count_within_interval = count_within_interval + 1;
    end
end


fprintf("\n# of times Confidence Interval contained true mean:\n");
fprintf('=================================================\n');
fprintf("%d out of 1000 times\n", count_within_interval);




% d) Plot three curves in a single figure corresponding to part c
```

```
%  - the first curve should show the lower limit of the confidence interval
%  - the second curve should show the upper limit of the confidence interval
%  - the third curve is the constant true mean value
%
% Note: All the curves should be plotted against the experiment repetition
% number, 1, 2, ..., 1000.
%
% Can you visually inspect the curves to report how many times the
confidence interval
% does not contain the true mean value?
% ========================================================================


figure;
plot(1:num_trials, lower_limits, 'b'); hold on;
plot(1:num_trials, upper_limits, 'r');
plot(1:num_trials, true_mean * ones(num_trials,1), 'k--');
set(gca, 'Color', 'w');
legend('Lower Confidence Interval', 'Upper Confidence Interval', 'True
Mean');
xlabel('Trial Number');
ylabel('Resistance (Ohm)');
title('Confidence Intervals over 1000 Trials');
grid on;


% e) Repeat parts b, c, and d if the manager tells you that the true mean
% resistor is 220 ohm
% ========================================================================

true_mean = 220;

% Repeat 1000 trials
count_within_interval = 0;
lower_limits = zeros(num_trials,1);
upper_limits = zeros(num_trials,1);

for i = 1:num_trials
    sample = true_mean + std_dev * randn(n,1);
    sample_mean = mean(sample);

    lower = sample_mean - margin_of_error;
    upper = sample_mean + margin_of_error;

    lower_limits(i) = lower;
    upper_limits(i) = upper;

    if (true_mean >= lower && true_mean <= upper)
        count_within_interval = count_within_interval + 1;
    end
end

fprintf("\n# of times Confidence Interval contained true mean (220 Ohm):\n");
fprintf('============================================================\n');
```

```
fprintf("%d out of 1000 times\n", count_within_interval);

% Plot the Confidence Interval curves for true mean = 220
figure;
plot(1:num_trials, lower_limits, 'b'); hold on;
plot(1:num_trials, upper_limits, 'r');
plot(1:num_trials, true_mean * ones(num_trials,1), 'k--');
set(gca, 'Color', 'w');
legend('Lower Confidence Interval', 'Upper Confidence Interval', 'True
Mean');
xlabel('Trial Number');
ylabel('Resistance (Ohm)');
title('Confidence Intervals over 1000 Trials (True Mean = 220 Ohm)');
grid on;
```

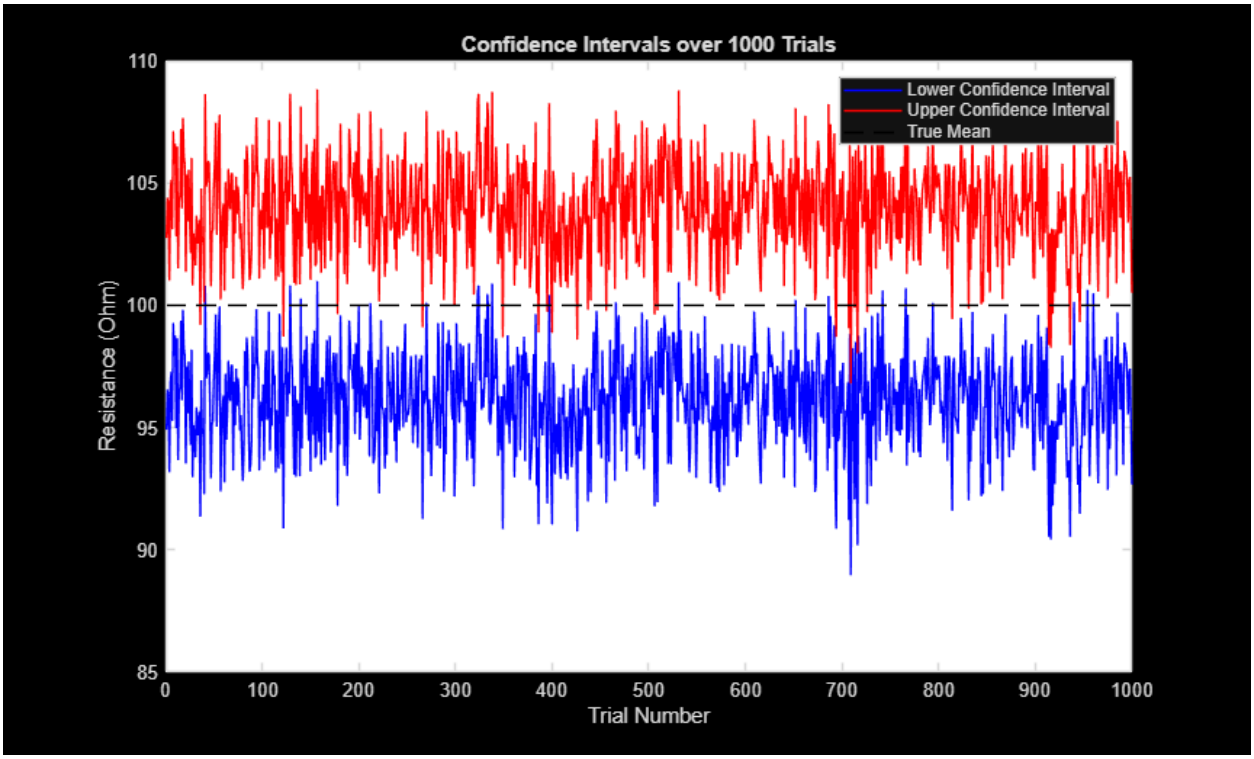
```
========================================
II) Design of an experiment:
========================================

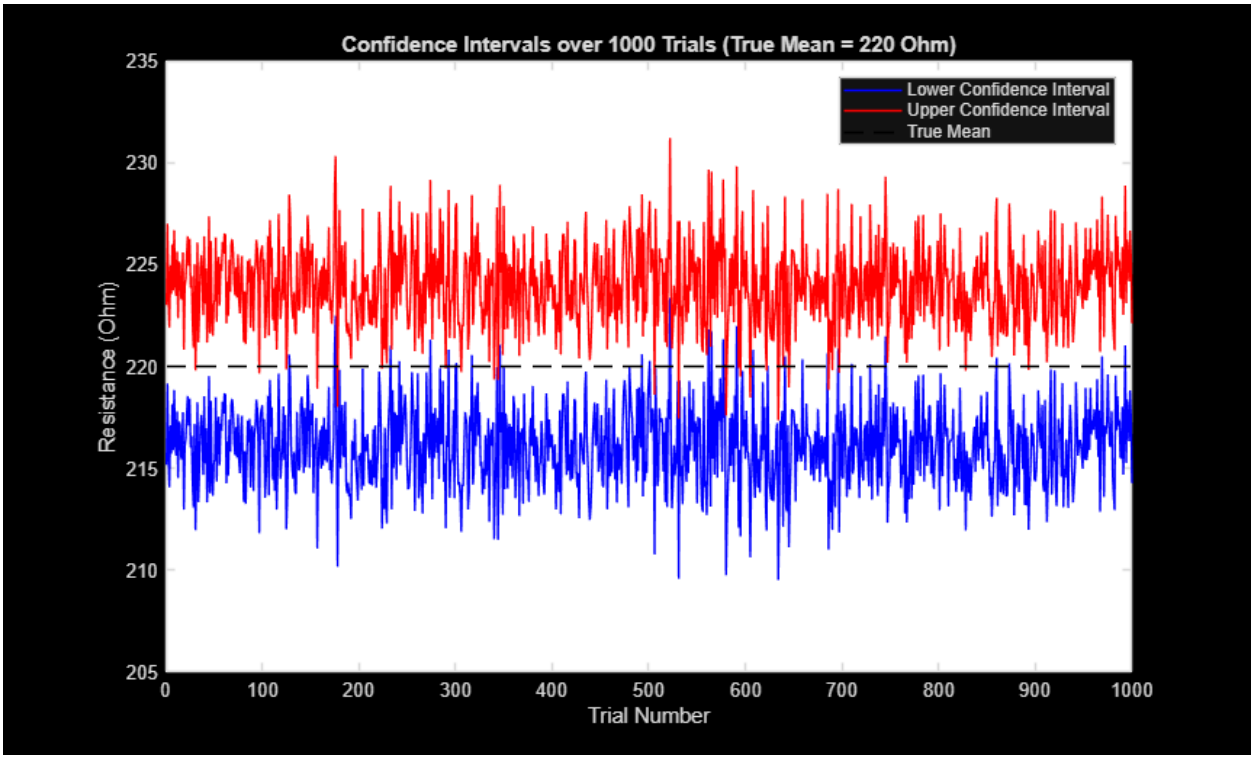
Samples needed to estimate the mean value:
==================================
25

Confidence Interval:
==================================
[95.48, 103.32]
Confidence Interval contains true mean

# of times Confidence Interval contained true mean:
=================================================
952 out of 1000 times

# of times Confidence Interval contained true mean (220 Ohm):
============================================================
943 out of 1000 times
```





Published with MATLAB® R2025b